6 Años: cartels game

Creation of a game based on the work of Rafael Prieto-Curiel
"Reducing cartel recruitment is the only way to lower violence in Mexico"

28/08/2026

KEYMESSAGES OF THE PUBLICATION

→ cartel in Mexico is **huge** (5th employer of the country)
→ **4 parameters** influencing the size of the cartel:
- recruitment
- incapacitation (= arrestation)
- ~~saturation (= retirement)~~
- ~~conflicts among cartel~~

→ **giving up = going worse**
→ **invest money** in arrestations and rehabilitation could make a difference

RESEARCH

CARTELS

Reducing cartel recruitment is the only way to lower violence in Mexico

Rafael Prieto-Curiel[1]*, Gian Maria Campedelli[2]†, Alejandro Hope[3]‡

Mexican cartels lose many members as a result of conflict with other cartels and incarcerations. Yet, despite their losses, cartels manage to increase violence for years. We address this puzzle by leveraging data on homicides, missing persons, and incarcerations in Mexico for the past decade along with information on cartel interactions. We model recruitment, state incapacitation, conflict, and saturation as sources of cartel size variation. Results show that by 2022, cartels counted 160,000 to 185,000 units, becoming one of the country's top employers. Recruiting between 350 and 370 people per week is essential to avoid their collapse because of aggregate losses. Furthermore, we show that increasing incapacitation would increase both homicides and cartel members. Conversely, reducing recruitment could substantially curtail violence and lower cartel size.

Latin America is home to only 8% of the world's population, but roughly one in three intentional homicides worldwide stability through extensive acts of violence (*8*, *9*). Furthermore, some cartels have acquired a transnational dimension, expanding their business to the United States and beyond (*10*).

In this context, although cartels lose dozens of members daily as a result of killings and state incapacitation through incarcerations, the violence over the years has not decreased. We tackle this puzzle by studying cartels' evolution, deriving their sizes, and considering four fundamental sources of size variation: recruitment, incapacitation, conflict, and saturation. These sources capture the different exogenous and endogenous dynamics explaining why and to what extent cartels grow or shrink. Recruitment refers to the process of attracting a new workforce that stably carries out tasks (both strictly criminal and not) for cartels (*11*). Incapacitation measures the ability of the state to counter cartels through incarcerations that occurred in the country (*16*). Although homicide and incarceration trends are imperfect because they do not discriminate between offenses that occurred specifically in the context of organized crime, they can be used to estimate cartels' violence capacity and the state's incapacitation against them. In this work, we build on this intuition and exploit data on murders, missing persons, and incarcerations in Mexico between 2012 and 2022 to derive cartel size. We propose a mathematical system to represent their behavior over 10 years and seek to shed light on the mechanisms within the so-called black box of the cartels.

This work has two main goals. First, we aim to obtain plausible estimates of the cartels' population, including their number of members and recruitment capacity. Second, we seek to simulate different policy scenarios (i.e., increased state incapacitation and recruitment prevention) to disentangle the effects of varying strategies to curb cartels' power and, in cartels' actions, as well as the insufficient amount of information accessible to map them, makes them conceptually similar to black boxes, from which we can only extrapolate imperfect proxies of activity using, for instance, the daily number of homicides or the number of drug-related policy strategy (focused on increasing police efforts to incarcerate cartel members)?

We consider four mechanisms that explain why cartel size varies: recruitment, incapacitation, saturation, and conflict (Fig. 1). We model the conflict between cartels with a

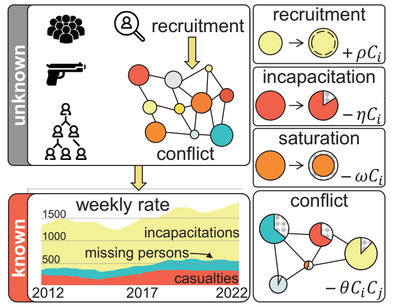


Fig. 1. Model diagram representing the four reasons why a cartel changes in size. Most cartel-related activities remain undercover, but we observe some of their by-products in casualties and incapacitations.

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RESEARCH

→ selection of the **subject**
→ deciding the **format**:
- game ?
- simulation ?
- physical or digital?

→ thinking about the **game dynamics**:
- not too far from the mathematic model
- but also fun to play

→ what kind of **game board** ? (map of Mexico, timeline)
→ how many **teams** and should play the cartel?



MOODBOARD

→ dark minimalist style (short timing, dark subject)
→ 1 animal per role (police=eagle social worker=hummingbird or butterfly) inspired by Rufino Tamayo art
→ square style inspired by Gunther Gerzso art
→ old daily journal style



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RISTS DESTROY WORLD TRADE CE
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What's News—

Business and Finance

ALL MAJOR U.S. FINANCIAL markets closed yesterday and remain closed today in the wake of the terrorist attack on the World Trade Center. The near-panic reaction in the global markets that remained open suggested that substantial damage was done to the psyche of a world financial system already on edge from prospects of an international recession. In Tokyo, the Nikkei stock index fell below 10000 early Wednesday for the first time since 1984. (Article on Page B1)

The World Trade Center housed many Wall Street and banking firms, law offices, technology companies, trading firms and other businesses. Many escaped before the destruction of the buildings yesterday, but the toll of dead and injured is unclear. (Article on Page B1)

The attacks threaten to push a fragile global economy into widespread recession, smashing consumer confidence and disrupting basic commercial functions such as air travel. (Article on Page A1)

The dollar tumbled in global markets following the attacks. In late London trading, the euro stood at $1.44 U.S. cents, up from 88.86 cents late Monday in New York. The dollar tumbled to 118.16 yen from 120.55 yen and the British pound rose to $1.4751 from $1.4579.

World-Wide

BUSH PROMISED action against terrorist attacks in the Eastern U.S.

The death toll from the hijacked-jet attacks that destroyed the World Trade Center's towers in New York and damaged the Pentagon outside Washington was impossible to gauge immediately. But the president said "thousands of lives were suddenly ended." A fourth hijacked plane crashed near Pittsburgh. Another commercial jet went down in western Pennsylvania. It wasn't immediately clear who was responsible for the attacks, but the president told the nation the U.S. "would make no distinction" between terrorists and "those who harbored them." He virtually promised armed response earlier yesterday, saying that "the United States will hunt down and punish those responsible for these cowardly acts," he said. (Articles on pages A1 and A13)

Sen. McCain, a Vietnam War veteran, expressed the incidents' gravity: "These were not just crimes against the United States, they are acts of war."

HEALTH TEAMS launched efforts to treat thousands of injured victims.

Health workers mobilized a nationwide effort to treat the thousands of injured taken to hospitals, identify the dead and supply tens of thousands of units of blood in the wake of the terrorist attacks. More than 2,600 people were reported injured in New York City, and hospitals expected to see more. Blood-center officials said immediate needs would be met by available supplies, but they worried that they would run short in coming days as they face the need to replenish supplies. (Article on Page A6)

Death Toll, Source of Devastating Attacks Re
U.S. Vows Retaliation as Attention Focuses on

By David S. Cloud And Neil King
Staff Reporters of The Wall Street Journal

By successfully attacking the most prominent symbols of American power—Wall Street and the Pentagon—terrorists have wiped out any remaining illusions that America is safe from mass organized violence.

That realization alone will alter the way the U.S. approaches its role in the world, as well as the way Americans travel and do business at home and abroad.

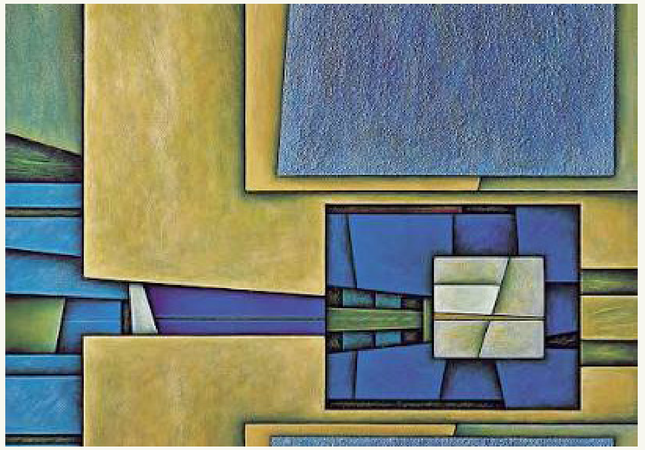
The death toll from the hijacked jets' attacks that destroyed the World Trade Center in lower Manhattan, and damaged the Pentagon, was impossible to gauge immediately. But it could eclipse the loss of life the country suffered in the Japanese attack on Pearl Harbor, when more than 2,300 perished.

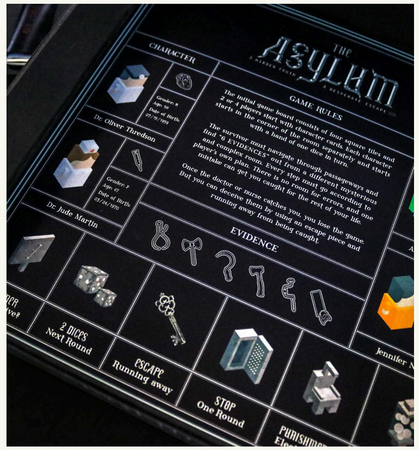
It wasn't immediately clear who was responsible for the attacks, but the president's attention focused on Middle East terrorist Osama bin Laden and his organization. One U.S. official said intelligence agencies already had gathered "strong information" linking Mr. bin Laden to the attacks. If the bin Laden organization isn't directly responsible, U.S. officials suspect, it could have sprung from a network of Islamic terror groups he supports and finances.

The gravity of the challenge to the country was summarized by Sen. John McCain, a Vietnam War veteran, who said: "These were not just crimes against the United States, they are acts of war."

Yet a war against terrorism is unlike a conventional war, and in some ways is far scarier. As a traumatized nation saw in gruesome detail on its television sets, terrorists attack civilians, not soldiers. And while the wars of the past century involved nation-states that could ultimately be defeated, a war against terrorism involves a less distinct enemy, whose de-







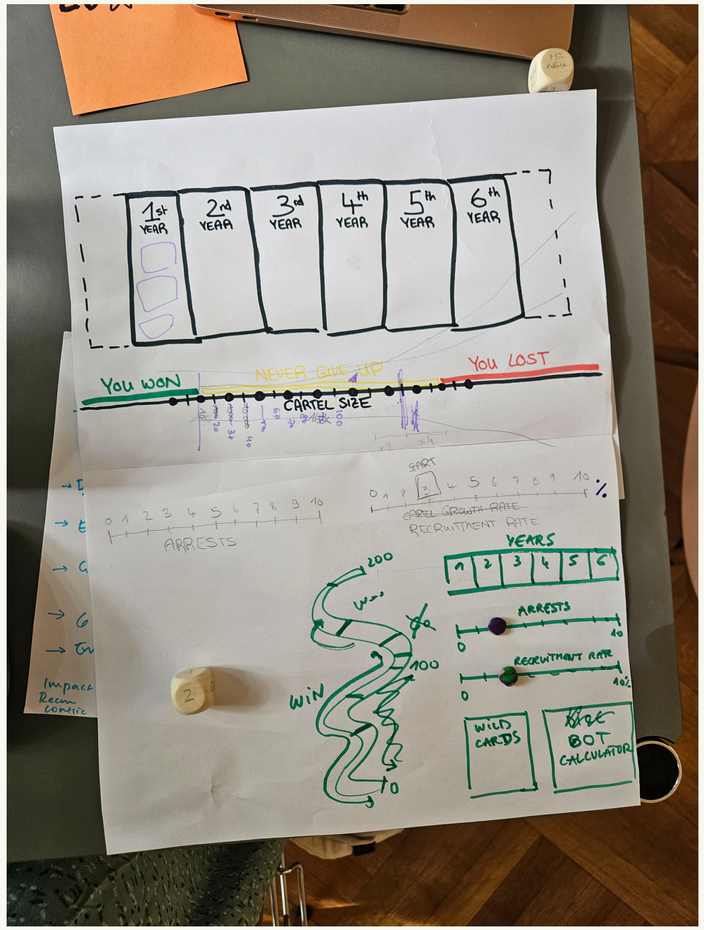


THE GAME GOAL

CONTEXT: The mexican government is hiring you to reduce the impact of the cartel on their population, you have six years to act before the government is changed.
GOAL: Reduce the cartel size.
RULES:
→ 2 teams:
- **Guardia Nacional (GN)**= national police
- **Non-governmental organization (NGO)** = social workers

→ Players from the same team have to work together
→ 6 years: 1 round per year
→ Each round: each player **roll a dice** (0-3) determining their **budget**. They can:
- **invest** for arrestments (GN) or rehabilitation (NGO)
- **save** for their personal interest

→ Each action has an impact on the size of the cartel
→ **Wildcards**: events influencing the different parameters.
TO WIN: cartel size <90. But also the team with the most money wins!



GAME COMPONENTS

→ A cartel track, 0 to 200, marked in tens, with four band lines at 40, 80, 120 and 160
→ A cartel marker showing the current cartel size on the track
→ A recruitment track, -10% to +10%, marked in single points
→ A recrutment marker showing the current recrutment rate
→ An arrest track, 0 to 10 arrests
→ A arrest marker showing the decided arrest numbers for that year
→ One die per team, faces 0 1 1 2 2 3
→ Coins in units of 100 pesos
→ 10 wildcards
→ A year counter, 1 to 6
→ A calcualtor with an app to imput the numbers each round



SETUP, STEP BY STEP

→ Split into two teams of equal size: Guardia Nacional and the NGO. Any size from 1 to 4 each.
→ Put the cartel marker on 100.
→ Put the recruitment marker on +3%.
→ Put the arrest marker at 0
→ Set the year counter to 1.
→ Give each team one die.
→ Shuffle the 10 wildcards face down. You will use six, one per year.
→ Both teams start with an empty bank.
→ Team’s action price: 100 pesos per player on your team. (e.g three players means every action costs 300).



A YEAR, STEP BY STEP

1.

Flip the wildcard.

Read it aloud and apply it now, before anyone rolls.

2.

Check the band. Look at where the cartel marker sits and move the recruitment marker accordingly.

Cartel is at	Move recruitment
0 to 40	+1%
41 to 80	+2%
81 to 120	+3%
121 to 160	+4%
161 to 200	+5%

3.

Everyone rolls.

Each player rolls their die once. Your team's coins for this year are the sum of your team's dice.
One coin = 100 pesos.



A YEAR, STEP BY STEP

4.

Buy, then bank.

Your team decides together how much to spend and how much to bank. Every 100 pesos per player buys one action. (e.g if you have two players you need 200 pesos per action)

Guardia Nacional buys arrests.

Each arrest moves the arrest marker up one space . Maximum 5 arrests in one year.

The NGO buys prevention.

Each prevention moves the recruitment marker down 1.

Everything you do not spend goes into your team’s bank. It stays there. You cannot take it back out.



A YEAR, STEP BY STEP

5.

The cartel grows.

Apply recruitment first, then arrests.

New cartel size = current size + (current size × recruitment % − number of arrests)rounded

6.

Advance the year. After year 6, score.



WINNING

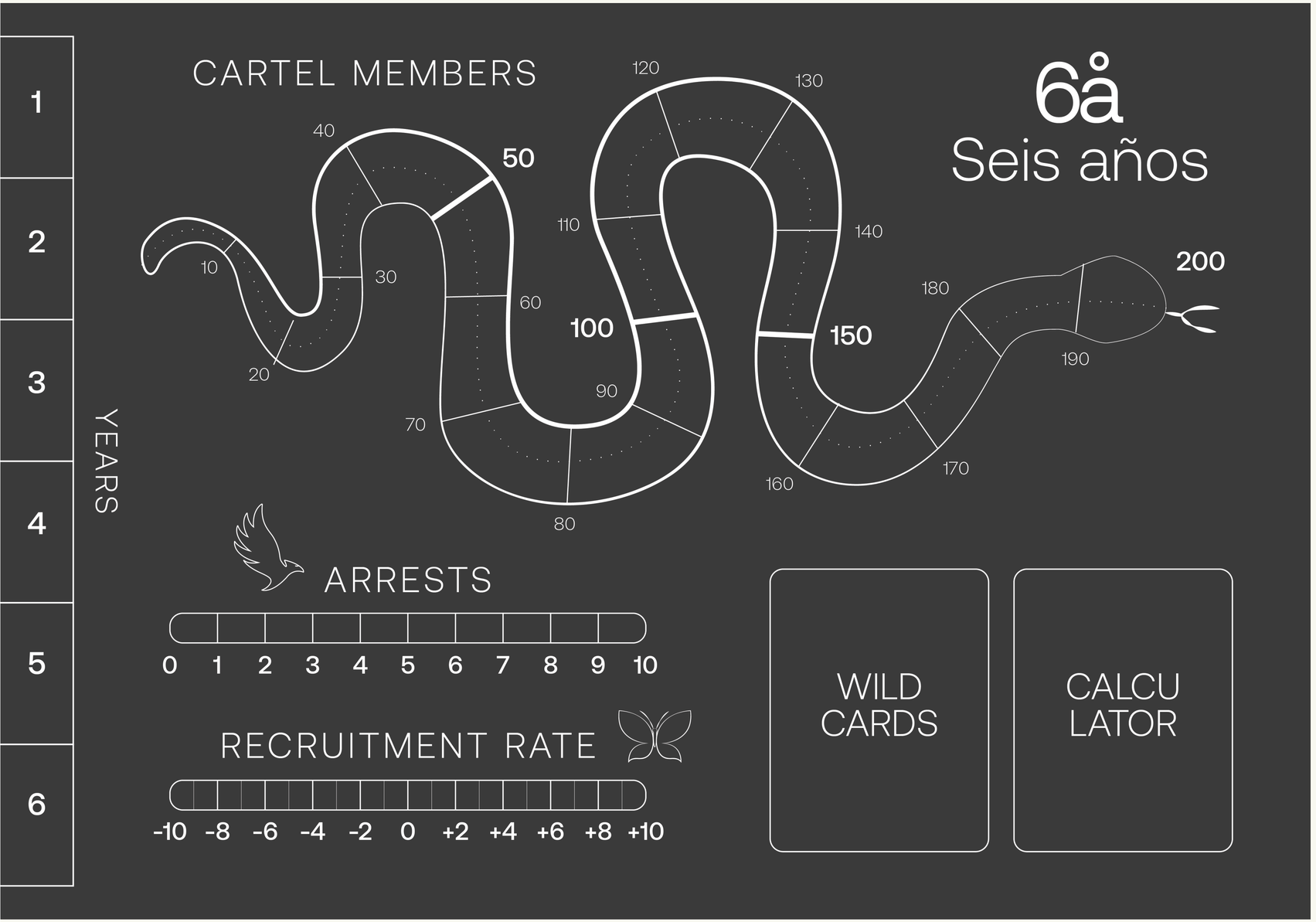
Both teams lose if the cartel is above 90.

If cartel loses, the team with more money in the bank wins the internal race. The Guardia gets its budget increase, or NGO gets its programme expansion. Ties go to nobody.

You cannot win the money race if the cartel wins. Hoarding is only worth something if somebody paid to beat the cartel.

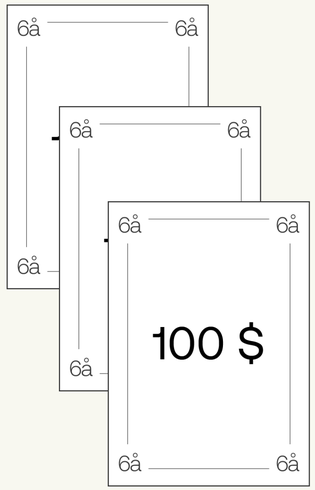


THE BOARD & CARDS





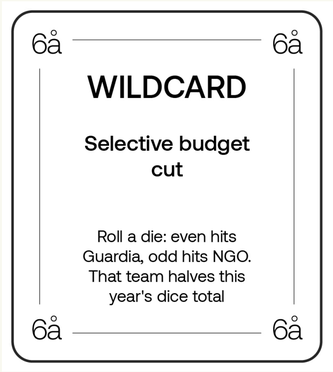


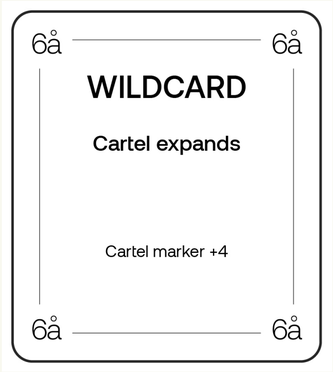




THE 10 WILDCARDS



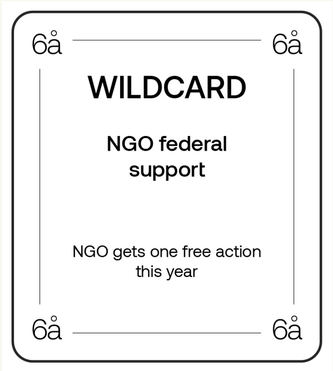


















TESTING THE GAME MECHANICS

PLAYTEST TABLE · ITERATION 0.3 · TWO VICTORIES

Seis Años

Six budget years. One cartel that recruits faster than you can arrest. The table wins together at 80 members or fewer. Then one of you wins alone.

Play a mandate | **Balance lab**

PARAMETERS

BUDGET CARD FLOOR	BUDGET CARD CEILING	PREVENTION CAP / YEAR	RECRUITMENT FLOOR	COST PER 1 PP	WIN THRESHOLD
$800	$1600	4 pp	4%	$200	80

DIVERSION APPETITE	SAFETY MARGIN KEPT	EVENT DECK
100%	0 members	Harsh (B)

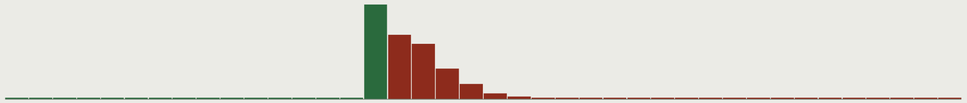
Run 4,000 mandates per strategy

Each strategy plays the same rules against fresh budget and event draws. Skilled play buys prevention up to the cap in the first half of the mandate, waits out corruption years, then converts the rest to arrests. Diversion appetite is how much of the money the table believes it can spare gets moved into personal funds. The safety margin is how far under the target players insist on staying before they take anything.

4,000 MANDATES EACH · BUDGET $800–$1600, CAP 4PP, FLOOR 4%, $200/PP, TARGET 80, DIVERTING 100% OF SLACK WITH A 0-MEMBER MARGIN

STRATEGY	TEAM WIN	DIVERTED	P10	MEDIAN	P90	
Skilled: prevention first		35.3%	$1586	79	83	94
Even split every year		0.7%	$1151	84	91	104
Random allocation		9.5%	$1161	80	91	110
Arrests only		19.3%	$737	79	90	112

FINAL CARTEL SIZE UNDER SKILLED PLAY · GREEN IS A WIN



PLAYTEST TABLE · ITERATION 0.3 · TWO VICTORIES

Seis Años

Six budget years. One cartel that recruits faster than you can arrest. The table wins together at 80 members or fewer. Then one of you wins alone.

Play a mandate | **Balance lab**

PARAMETERS

BUDGET CARD FLOOR	BUDGET CARD CEILING	PREVENTION CAP / YEAR	RECRUITMENT FLOOR	COST PER 1 PP	WIN THRESHOLD
$800	$1100	8 pp	1%	$350	95

DIVERSION APPETITE	SAFETY MARGIN KEPT	EVENT DECK
100%	0 members	Harsh (B)

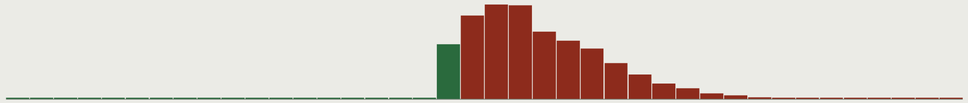
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4,000 MANDATES EACH · BUDGET $800–$1100, CAP 8PP, FLOOR 1%, $350/PP, TARGET 95, DIVERTING 100% OF SLACK WITH A 0-MEMBER MARGIN

STRATEGY	TEAM WIN	DIVERTED	P10	MEDIAN	P90	
Skilled: prevention first		9.1%	$506	95	109	130
Even split every year		1.3%	$420	100	115	137
Random allocation		1.4%	$454	103	121	143
Arrests only		12.6%	$426	95	109	131

FINAL CARTEL SIZE UNDER SKILLED PLAY · GREEN IS A WIN





Thank you